PHANTOM 4.8 // DIAGRAM 11 — 80,000-CAMERA STATE-WIDE SCALABILITY ARCHITECTURE

GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2 EXHIBIT

GUJARAT STATE-WIDE SURVEILLANCE SCALE // EXECUTIVE COMPUTATION MATRIX

80,000
CCTV CAMERAS
Across 33 Gujarat Districts

200 Gbps
RAW VIDEO STREAM
80K x 2.5 Mbps uncompressed

99.92%
BANDWIDTH REDUCTION
WAN load drops to 154 Mbps

1,667
NVIDIA L40S GPUs
~50.5 GPUs per District CCC

TIER 1: 33 DISTRICT EDGE NETRAM COMMAND CLUSTERS (EDGE INFERENCE)

Ahmedabad Netram CCC
- Ingests 25,000 CCTV feeds
- 520x NVIDIA L40S GPUs
- 1-Slot decoupled frame buffers

Local YOLO26 & ANPR inference
Metadata only pushed upstream

Surat Netram CCC
- Ingests 18,000 CCTV feeds
- 375x NVIDIA L40S GPUs
- Local incident state machine

Sub-25ms glass-to-detection
District FOV wedge spatial map

31 Other District CCCs
- Vadodara, Rajkot, Bhavnagar...
- 772x GPUs distributed
- Autonomous air-gapped run

Survives central WAN severed
Local Redis buffer (10K alerts)

METADATA ONLY: 153.6 Mbps

TIER 2: REGIONAL ZONE GATEWAYS (APACHE KAFKA & REDIS CLUSTERS)

North & Central Zone Gateway (Ahmedabad)
- 5-Broker Apache Kafka Cluster (cctv.events topic)
- Redis Sentinel hot cache for cross-district plate match

Aggregates 45,000 camera event streams

South & Saurashtra Zone Gateway (Surat)
- 5-Broker Apache Kafka Cluster (cctv.events topic)
- Redis Sentinel hot cache for coastal & highway alert bus

Aggregates 35,000 camera event streams

TIER 3: CENTRAL STATE COMMAND & CONTROL CENTER (GANDHINAGAR CCC)

Master Database Grid
- PostgreSQL 16 + PostGIS
- Patroni HA (3-Node cluster)
- Temporal table partitioning
- 80,000 camera GIS index

Sub-10ms spatial queries

Global Watchlist sync

State AI Copilot & BI
- SurveillanceCopilot Agent
- 40+ Grounded tools active
- Multilingual NLP (EN, HI, GU)
- Cross-district trajectory solver

High-level threat reasoning

Certified dossier builder

Command HUD & Dispatch
- State-wide executive wall
- WebSocket alert multiplexer
- Cross-district PCR dispatch
- Incident FSM master sync

Zero duplicate alerts

Real-time threat interception

TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- Edge Processing: 33 District Netram CCC Edge Clusters process video locally; only JSON metadata flows over WAN.
- Bandwidth Reduction: Reduces WAN bandwidth from 200 Gbps (raw video) to ~154 Mbps (metadata only) — 99.92% savings.
- Compute Sizing: 1,667 NVIDIA L40S GPUs (avg 50.5 per district) batch-processing 48 streams each @ 15 FPS.